

Network Traffic Analysis of Ransomware Activity in the Healthcare Sector Using Wireshark and MITRE ATT&CK

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According to the recent Internet crime report by the US Federal Bureau of Investigation, financial losses in the healthcare and public Health sections constituted \$206 Million USD in 2024. Ransomware attacks played a key role. Usually, prior to deploying ransomware in a system, the attacker would use different tactics at various stages to achieve their tactical goals. The MITRE ATT&CK framework provides a knowledge base of tactics and techniques of real world threat actors. The tactics include reconnaissance, initial access, credential access, privilege escalation, and data exfiltration

This study applied the MITRE ATT&CK framework combined with network traffic analysis to examine a ransomware attack within a healthcare network. The study was done on an existing packet capture. The results of the analysis revealed multiple IoCs including ransomware activity, communication with external IP addresses, encrypted outbound traffic potentially associated with command-and-control channels, and lateral movement between internal hosts. Unauthorized access and file transfer activities were confirmed from cleartext credentials, exposure via FTP.

The findings from this study demonstrate how network traffic analysis, when aligned with MITRE ATT&CK, can support systematic identification and understanding of adversary behavior. This work demonstrates practical cybersecurity analysis by showing how observable network artifacts can be used to understand attack progression. Future work may explore automated identification of IoCs and plans to improve monitoring of insecure protocols in healthcare environments.